Algorithm

Step 1: Start the program.
Step 2: initialize the FIFO Queue pointers front and rear to NULL and set the initial patient token number to 1001.
Step 3: initialize the doubly linked list pointers head and tail to NULL
Step 4: Display the main menu with the following operations
1) Register patient
2) call next patient
3) Search patient
4) Display All patient Records
5) Display records in Reverse
6) count waiting Patients
7) Hospital Dashboard
8) discharge patient
9) Exit.

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Step 5 :- Accept the user's choice
Step 6 :- Register patient
1) Create a new patient node dynamically.
2) Enter the patient's name, age and department.
3) Generate a unique token number and initialize the patient - status to waiting.
4) insert the patient at the end of doubly linked list by updating the prev, Record and next pointers.
5) insert at patient at the rear of the fifo Queue using the next Queue pointer.
6) Display the token number, queue position and estimated waiting time

Step 7 :- Search patient
1) Accept the patient's token number
2) Start traversing the Doubly linked list from head.
3) compare the token number of each patient with the entered token.
4) if a match is found, display the patient's details and current status.
5) if no match is found, display "patient's not found".

Step 16

Step 12: Display hospital dashboard
1) initialize the total, waiting, served, and discharged counters to zero.
2) Traverse the doubly linked list.
3) count all registered patient and classify them according to their status.
4) Display the total number of registered waiting, served and discharged patients
5) Display the estimated queue waiting time

Step 13: Discharge patient.
1) Accep the patient's token number
2) search for the patient in the doubly linked list.
3) if the patient is not found, display "Patient not found"
4) if the patient is still waiting do not allow discharge.
5) otherwise discharge patient.

Step 14: if exit is selected traverse the doubly linked list and release all dynamically allocated memory.

Step 15: Display the program termination message

Step 16: Stop the program.

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